AS · Edexcel · Physics

6 mins · 6 questions

Multiple Choice Questions

Refraction, Reflection & Polarisation

Equation for the Intensity of Radiation / Refraction & Refractive Index / Critical Angle / Total Internal Reflection / Measuring Refractive Index / Converging & Diverging Lenses / Using Ray Diagrams / Power of a Lens / Thin Lenses in Combination / Real & Virtual Images / The Lens Equation / Magnification / Plane Polarisation

Scan here to return to the course
or visit savemyexams.com

Total Marks **/6**

1 Which row of the table shows a base quantity and its base SI unit?

	Quantity	**Unit**
A	charge	C
B	length	m
C	mass	g
D	temperature	°C

(1 mark)

2 An object is placed in front of a lens.

Which row of the table shows a combination that will produce a real image of the object?

	Focal length of lens/ cm	**Object distance / cm**
A	−5	10
B	−5	2
C	5	10
D	5	2

(1 mark)

3 A system of lenses consists of a converging lens and a diverging lens in contact.

The magnitude of the power of the converging lens is 9.4 D and the magnitude of the power of the diverging lens is 4.2 D.

Which of the following is the power of this system of lenses?

A. 13.6 D

B. 5.2 D

C. –5.2 D

D. –13.6 D

(1 mark)

4 For total internal reflection to take place, the angle of incidence must be

A. greater than or equal to the critical angle.

B. greater than the critical angle.

C. less than or equal to the critical angle.

D. less than the critical angle.

(1 mark)

5 An object is placed 6.5 cm from a lens of focal length 3.9 cm. An image is formed 9.8 cm behind the lens.

Which of the following expressions is equal to the magnification?

A. $\frac{3.9}{6.5}$

B. $\frac{6.5}{9.8}$

C. $\frac{6.5}{3.9}$

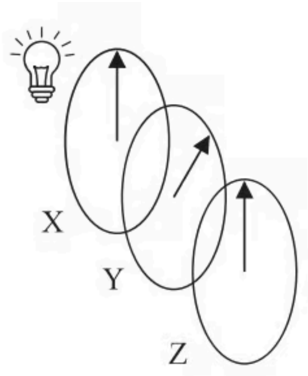
D. $\frac{9.8}{6.5}$

(1 mark)

6 Three polarising filters X, Y and Z, are placed in front of a source of unpolarised light.

The planes of polarisation of the filters are initially parallel.

Filter Y is rotated by 45° as shown.



Filter Z is then rotated clockwise and the intensity of light emerging from Z is measured.

Which angle of rotation of Z will result in the lowest intensity of light?

A. 90°

B. 135°

C. 180°

D. 225°

(1 mark)